



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2004

MATHEMATICS EXTENSION 1

**1:25pm – 3:30 pm
Thursday 19th August 2004**

Directions to Students

- Reading Time : 5 minutes
- Working Time : 2 hours
- Write using blue or black pen. (sketches in pencil).
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.
- **Answer each question in the booklets provided and clearly label your name and teacher's name.**
- Total Marks 84
- Attempt Question 1 – 7
- All questions are of equal value

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Total marks (84)

Attempt Questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 Marks) Use a SEPARATE writing booklet. Marks

(a) Solve $\frac{5}{2x-1} < 3$. 3

(b) Find the acute angle between the lines $2x - y + 1 = 0$ and $x + 3y - 4 = 0$. 3
Give answer to the nearest degree.

(c) Find the coordinates of the point that divides the interval joining $(-2, 5)$ and $(8, -9)$ internally in the ratio $2 : 3$. 2

(d) If α, β, γ are the roots of the equation $x^3 - 5x^2 - 3x + 2 = 0$, 2
find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$.

(e) Write down the general solution, in terms of π , of the equation 2
$$\cos \theta = -\frac{1}{2}.$$

QUESTION 2 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Use the substitution $x = u^2 + 1$, for $u > 0$, to evaluate

4

$$\int_1^5 (x+1)\sqrt{x-1} \, dx.$$

- (b) Evaluate $\int_0^{\frac{\pi}{4}} \sin^2\left(\frac{1}{2}x\right) dx$.

3

- (c) Prove, using the principle of mathematical induction, that $9^{n+2} - 4^n$ is divisible by 5, for n a positive integer.

5

QUESTION 3 (12 Marks) Use a SEPARATE writing booklet.

Marks

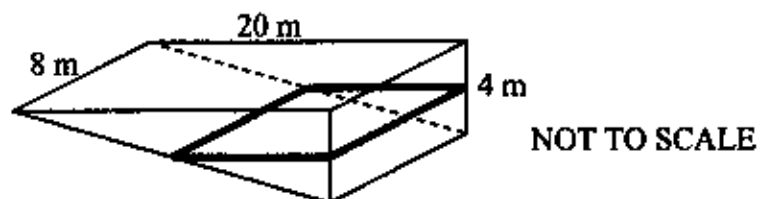
- (a) Find the exact value of $\tan\left(2\sin^{-1}\frac{3}{4}\right)$. **3**
- (b) Consider the function $f(x) = \sin^{-1}(x+1) + \frac{\pi}{2}$.
- (i) What is the domain of $f(x)$? **1**
- (ii) Sketch the graph of $y = f(x)$. **2**
- (c) Consider the function $f(x) = \log_e(2x+1)$.
- (i) Write down the domain of $f(x)$. **1**
- (ii) Find the inverse function of $f(x)$, and write it in the form $f^{-1}(x) = \dots\dots$ **2**
- (iii) Find the gradients of the graphs of $y = f(x)$ and $y = f^{-1}(x)$ at the origin. **1**
- (iv) On the same diagrams, draw the graphs of $y = f(x)$ and $y = f^{-1}(x)$. **2**

QUESTION 4 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the coefficient of x^3 in the expansion of $(2-x)(1+x)^5$. 3

(b)



A swimming pool is 20 metres long, 8 metres wide, 4 metres deep at one end, and zero depth at the other end. The floor of the pool is a plane rectangular surface.

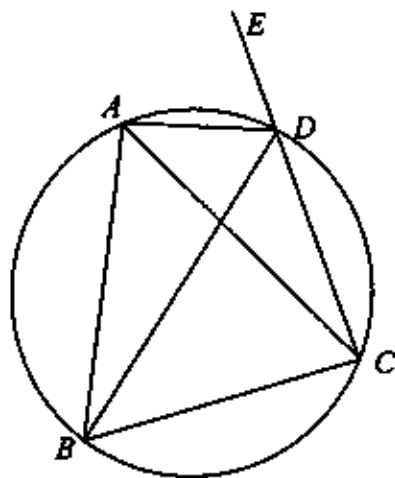
- (i) When the depth of water at the deeper end is h metres, show that the volume ($V \text{ m}^3$) of water in the pool is given by $V = 20h^2$. 2
- (ii) If water is being poured into the pool at the rate of $2 \text{ m}^3/\text{minute}$, find the rate at which the depth of the water is increasing at the deepest end, when the depth is 1 metre. 2
- (c) The value of a home business, $\$V$, is increasing at a rate proportional to the amount by which the value is less than $\$4000$.
i.e. $\frac{dV}{dt} = k(4000 - V)$
Initially, the value of the business was $\$2000$ and after 5 years it was $\$3000$.
- (i) Show that $V = 4000 - Ae^{-kt}$ satisfies this equation. 1
- (ii) Find the value of A and the value of k to 4 decimal places. 2
- (iii) Find the number of years for the value of the business to grow to $\$3800$. 2

QUESTION 5 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Show that the derivative of $x^2 e^{-x}$ is $xe^{-x}(2-x)$. 1
- (ii) Show that $x^2 e^{-x} = 0.4$ has a root between $x = 1$ and $x = 2$. 1
- (iii) Use Newton's approximation to find an approximation to the root of $x^2 e^{-x} = 0.4$, taking $x = 1$ as a first approximation. 3

(b)



$ABCD$ is a cyclic quadrilateral in which $AB = AC$, and CD is produced to E .
Prove that AD bisects the angle BDE .

3

- (c) In the expansion of $(3+2x)^8$, c_r is the coefficient of x^r .

- (i) Show that $\frac{c_r}{c_{r-1}} = \frac{18-2r}{3r}$. 2
- (ii) Hence or otherwise find the largest coefficient in the expansion of $(3+2x)^8$. 2

QUESTION 6 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) The position of a particle at time t is given by:

$$x = 3 \sin 2t - 4 \cos 2t.$$

- | | | |
|-------|---|---|
| (i) | Show that this equation satisfies $\ddot{x} = -n^2 x$. | 2 |
| (ii) | What is the initial velocity of the particle? | 1 |
| (iii) | At what time does the particle first come to rest? | 3 |

- (b) The acceleration of a particle at position x is given by:

$$\ddot{x} = -\frac{1}{4x^3}.$$

Initially the particle is at $x = 1$ moving with a velocity of $\frac{1}{2}$ unit in the positive direction.

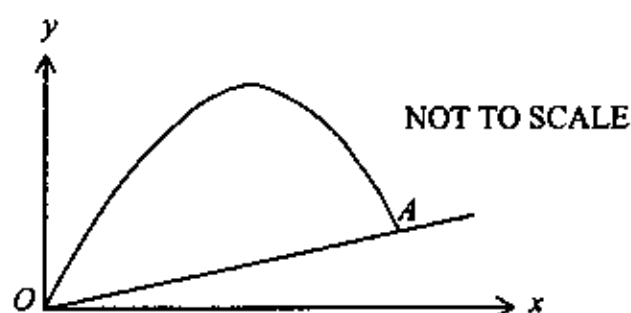
- | | | |
|-----|--|---|
| (i) | Prove that the velocity of the particle at position x is given by: | 3 |
|-----|--|---|

$$v = \frac{1}{2x}.$$

- | | | |
|------|---|---|
| (ii) | Hence find the position of the particle at time t . | 3 |
|------|---|---|

QUESTION 7 (12 Marks) Use a SEPARATE writing booklet.

Marks



An object is thrown from ground level with a speed of 40 m/s at an angle of 60° to the horizontal. Assume acceleration due to gravity is 10 m/s^2 and neglect air resistance.

- (a) Find equations for x and y in terms of time t seconds, starting from the acceleration equations $\ddot{x} = 0$ and $\ddot{y} = -10$, and hence show that: **4**

$$y = \sqrt{3}x - \frac{x^2}{80}.$$

- (b) The object is thrown up a slope with a gradient of $\frac{1}{4}$. **2**
Show that the horizontal distance travelled by the object when it lands on the slope is given by:

$$x = 80\sqrt{3} - 20.$$

- (c) Hence find the distance OA (to the nearest metre) up the slope from the point of projection to the point of landing. **2**

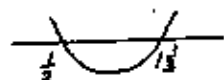
- (d) Show that the maximum height reached by the object above the slope is **4**
 $(61.25 - 10\sqrt{3})$ metres.

End of paper

(2004 Trials)

MATHEMATICS EXTENSION 1 - QUESTION 1

(a) $\frac{5}{2x-1} < 3$
 $5(2x-1) < 3(2x-1)^2$
 $3(2x-1)^2 - 5(2x-1) > 0$
 $(2x-1)[3(2x-1) - 5] > 0$
 $(2x-1)(6x-8) > 0$
 $x < \frac{1}{2} \text{ or } x > \frac{4}{3}$



③

(b) $2x - y + 1 = 0$ $m_1 = 2$
 $x + 3y - 4 = 0$ $m_2 = -\frac{1}{3}$
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{2 - (-\frac{1}{3})}{1 + 2(-\frac{1}{3})} \right|$
 $= \frac{2\frac{1}{3}}{\frac{1}{3}}$
 $= 7$
 $\theta = 82^\circ$

③

(c) $(-2, 5), (8, -9)$ $2:3$
 $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) = \left(\frac{2 \times 8 + 3 \times (-2)}{2+3}, \frac{2 \times (-9) + 3 \times 5}{2+3} \right)$
 $= \left(2, -\frac{3}{2} \right)$

②

(d) $x^3 - 5x^2 - 3x + 2 = 0$
 $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\gamma + \alpha + \beta}{\alpha\beta\gamma} = \frac{5}{-2} = -2.5$

②

(e) $\cos \theta = -\frac{1}{2}$
 $\theta = 2n\pi \pm \frac{2\pi}{3}$
 OR $\theta = (2n+1)\pi \pm \frac{\pi}{3}$
 OR equivalent.

②

Marks Awarded	Marker's Comments
(a). 1 mark 1 mark 1 mark	$5(2x-1) < 3(2x-1)^2$... or critical points. $2(2x-1)(3x-4) > 0$ $x < \frac{1}{2}, x > \frac{4}{3}$ or correctly solving the inequality obtained (unless trivial).
(b). 1 mark 1 mark 1 mark	$m_1 = 2 / m_2 = -\frac{1}{3}$. $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $... understanding this is formula to use even if not stated in this form. number crunching $\rightarrow 82^\circ$.
(c). 1 mark. 1 mark	$\frac{(2 \times 8) + 3(-2)}{2+3}$ $\frac{2 \times (-9) + 3 \times 5}{2+3}$ $(2, -\frac{3}{2})$
(d) 1 mark 1 mark	$\frac{\delta + \beta + \alpha}{\alpha\beta\gamma}$ $\alpha + \beta + \alpha = 5$ and $\alpha\beta\gamma = -2$.
(e) 1 mark. 1 mark	$2n\pi$ or equivalent $\pm \frac{2\pi}{3}$ or equivalent (must be $\pm$)

Question 2

$$2) \int_1^5 (x+1)\sqrt{x-1} dx$$

$$x = u^2 + 1$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du \quad [1]$$

$$x = 1 \quad u = 0$$

$$x = 5 \quad u = 2$$

$$= \int_0^2 (u^2 + 2)\sqrt{u} \cdot 2u du$$

$$= \int_0^2 (2u^4 + 4u^2) du \quad [2]$$

$$= \left[\frac{2u^5}{5} + \frac{4u^3}{3} \right]_0^2$$

$$= \left[\left(\frac{64}{5} + \frac{32}{3} \right) - (0) \right]$$

$$= \frac{352}{15} = 23\frac{7}{15} \quad [1]$$

$$b) \int_0^{\frac{\pi}{4}} \sin^2\left(\frac{1}{2}x\right) dx = \int_0^{\frac{\pi}{4}} \frac{1}{2}(1 - \cos x) dx \quad [1]$$

$$= \frac{1}{2} \left[x - \sin x \right]_0^{\frac{\pi}{4}} \quad [1]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{\sqrt{2}} \right] \quad [1]$$

c) Prove $9^{n+2} - 4^n$ is divisible by 5.

i) let $n=1$ $9^3 - 4 = 725 \therefore$ True for $n=1$ [1]

ii) Assume true for $n=k$ i.e. $9^{k+2} - 4^k = 5m$ (m is pos integer) [1]

$$\begin{aligned} \text{iii) When } n=k+1 \quad 9^{k+3} - 4^{k+1} &= 9(9^{k+2}) - 4(4^k) \\ &= 9(5m + 4^k) - 4(4^k) \\ &= 45m + 9 \cdot 4^k - 4 \cdot 4^k \\ &= 45m + 5 \cdot 4^k \\ &= 5[9m + 4^k] \end{aligned} \quad [2]$$

This is divisible by 5.

$\therefore$ if true for $n=k$, then true for $n=k+1$

$\therefore$ by induction, it is true for all $n \in \mathbb{N}$ [1]

Marks Awarded	Marker's Comments
<u>1</u>	Many mucked up the conversion.
<u>2</u>	Those with correct step one went on to get the correct integral and find the correct numeric value.
<u>1</u>	
<u>1</u>	Many did not know $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$
<u>1</u>	most integrated correctly.
<u>1</u>	most had correct evaluation.
<u>1</u>	
<u>1</u>	Almost all proved true for $n=1$.
<u>1</u>	$9^{k+2} - 4^k = 5m$ well stated. Most did not state that m was a positive integer.
<u>2</u>	Many could not set out the correct steps for this section. Many used "m" again. Is it the same "m" used earlier?
<u>1</u>	Many were lazy in their final statement. $\therefore$ most did not get this mark.

MATHEMATICS EXTENSION I - QUESTION 3

Mathematics Extension One: Question Number

3

Marker: Mr M'Car

(a) $\tan(2 \sin^{-1} \frac{3}{4})$

Let $\theta = \sin^{-1} \frac{3}{4} \therefore \sin \theta = \frac{3}{4}$

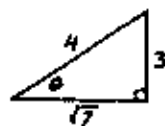
$\tan(2 \sin^{-1} \frac{3}{4}) = \tan 2\theta$

$= \frac{2 \tan \theta}{1 - \tan^2 \theta}$

$= \frac{2 \times \frac{3}{4}}{1 - \frac{9}{16}}$

$= -3\sqrt{7}$

③

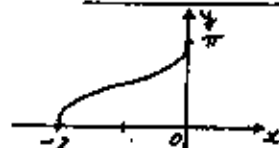


(b) $f(x) = \sin^{-1}(x+1) + \frac{\pi}{2}$

(i) Domain: $-1 \leq x+1 \leq 1 \therefore -2 \leq x \leq 0$

①

(ii)



②

(c) $f(x) = \log_e(2x+1)$

(i) Domain: $2x+1 > 0 \therefore x > -\frac{1}{2}$

①

(ii) $y = \log_e(2x+1)$

$2x+1 = e^y$

Inverse is: $2y+1 = e^x$

$y = \frac{1}{2}(e^x - 1)$

$f^{-1}(x) = \frac{1}{2}(e^x - 1)$

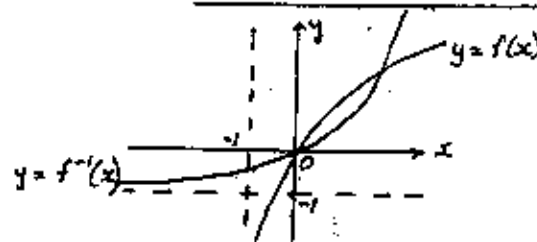
②

(iii) $f'(x) = \frac{2}{2x+1}$; $f'(0) = 2$

$\frac{d}{dx} f'(x) = \frac{1}{x^2}$; At $x=0$, $\frac{d}{dx} f'(x) = \frac{1}{x^2}$

①

(v)



②

(a)

Marks Awarded	Marker's Comments
1	$\tan \theta = \frac{3}{\sqrt{7}}$
1	double angle formula substit
1	correct final answer
	Note - 1 mark awarded for calculator answer ≈ 7.94
(b)	
(i) 1	Correct domain $-2 \leq x \leq 0$
(ii) 1	Correct shape
1	Correct position
	Note - Use a stencil !!
(c)	
(i) 1	Correct domain $x > -\frac{1}{2}$
(ii) 1	Interchange $x \leftrightarrow y$
1	Make y the subject
	(Generally well done)
(iii) 1	Either answer $f'(0) = 2$
	or $\frac{d}{dx} f'(0) = \frac{1}{x^2}$
(iv) 1	One correct function
or	
2	Must pass thru' origin } required for full marks
	Show both asymptotes }
	intersect twice }

(poorly answered)

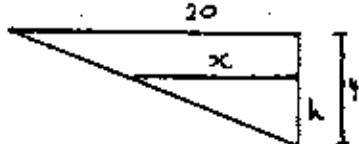
Question 4

$$2) (2-x)(1+x)^5 = (2-x)(1+5x+10x^2+10x^3+\dots)$$

$$= 2(10x^3) + (-x)(10x^2)$$

$$= 20x^3 - 10x^3$$

$$= 10x^3 \quad \therefore \text{Coefficient} = 10$$

i) 

$$\frac{h}{x} = \frac{4}{20} \quad V = \frac{1}{2} h x \times 8$$

$$20h = 4x \quad = \frac{1}{2} h \cdot 5h \cdot 8$$

$$x = 5h \quad = 20h^2$$

ii) $\frac{dV}{dt} = 2$ $h=1$ $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$

$$\frac{dV}{dh} = ? \quad 2 = 40h \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.05$$

iii) $\frac{dV}{dt} = k(4000 - V)$

i) $V = 4000 - Ae^{-kt}$

$$\frac{dV}{dt} = -k(-Ae^{-kt})$$

$$= +k(4000 - V)$$

ii) $t=0 \quad V=2000 \quad 2000 = 4000 - Ae^0$

$$A = 2000$$

$t=5 \quad V=3000 \quad 3000 = 4000 - 2000e^{-5k}$

$$e^{-5k} = \frac{1}{2}$$

$$k = -\frac{1}{5} \ln 0.5 = 0.1386$$

iii) $3800 = 4000 - 2000e^{-kt}$

$$e^{-kt} = 0.1$$

Marks Awarded	Marker's Comments
1	Correct expansion
1	Correct collection of coefficients
1	10. (Well done)
1	Correct explanation of why ratio was 5:1.
1	Correct explanation of why $V = 20h^2$. (Poorly done)
1	$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ or equivalent.
1	0.05 or $\frac{1}{20}$ or equivalent. (Well done)
1	Use of $Ae^{-kt} = 4000 - V$.
1	Evaluate A
1	Evaluate k (correct dp's)
1	Correct equation.
1	Correct value of t.

MATHEMATICS EXTENSION I - QUESTION 5.

- (a) $y = x^2 e^{-x}$
- (i) $\frac{dy}{dx} = e^{-x}(2x) + x^2(-e^{-x})$
 $= 2xe^{-x} - x^2 \cdot e^{-x}$
 $= xe^{-x}(2-x)$ ①
- (ii) Let $f(x) = x^2 e^{-x} - 0.4$: $f(1) = e^{-1} - 0.4 = -0.0360$
 $f(2) = 4e^{-2} - 0.4 = 0.19 > 0$
 Since $f(x)$ is continuous and $f(1), f(2)$ have opposite signs,
 a root lies between 1 and 2. ①
- (iii) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ $f'(1) = e^{-1} \times 1$
 $= 1 - \frac{-0.03}{0.3679}$ $= 0.3679$ ③
 $= 1.08$

- b) Let $\angle ADE = \theta$
- $\therefore \angle ABC = \theta$ (exterior angle cyclic quad)
 - $\therefore \angle ACB = \theta$ (base angles isosceles triangle)
 - $\therefore \angle ADB = \theta$ (angles in same segment)
 - $\therefore \angle ADE = \angle ADB$
 - $\therefore AD$ bisects $\angle BDE$.



- c) $(3+2x)^8$
- (i) $\frac{C_r}{C_{r+1}} = \frac{\binom{8}{r} 3^{8-r} 2^r}{\binom{8}{r+1} 3^{8-(r+1)} 2^{r+1}}$
 $= \frac{2}{3} \times \frac{8!}{r!(8-r)!} \times \frac{(r-1)!(9-r)!}{8!}$
 $= \frac{2}{3} \times \frac{9-r}{r}$
 $= \frac{18-2r}{3r}$ ②
- (ii) $\frac{18-2r}{3r} > 1$
 $18 > 5r$
 $r < 3\frac{3}{5}$
 Greatest coefficient = coefficient of x^3
 $= \binom{8}{3} 3^5 2^3$
 OR 108 864 ②

Mathematics Extension One: Question Number

5

Marker: Mr M

Marks Awarded	Marker's Comments
(a) (i) 1	Correct use of product rule
(ii) 1	Show a change of sign
(iii) 1	$f(1) = -0.03$
1	$f'(1) = 0.368$
1	Correct estimate $x_1 = 1.087$
	Note: must use $f(x) = x^2 e^{-x} - 0.4$
	Poorly answered by many students.
(b) 1	Exterior $\angle$ of cyclic quad.
1	Base $\angle$ s of isos. Δ
1	Angles in same segment.
	Note - very poor structure
	- drawing a diagram helps.
(c) (i) 1	$\frac{\binom{8}{r} 3^{8-r} 2^r}{\binom{8}{r+1} 3^{8-(r+1)} 2^{r+1}} \approx \frac{n-r+1}{r} \cdot \frac{b}{a}$
1	Use of factorial definition to correctly simplify.
(ii) 1	Solving inequality $r < 3\frac{3}{5}$
1	Finding greatest term 108 864
	Note: part (i) poorly answered.

Question 6

i) $x = 3 \sin 2t - 4 \cos 2t$

$\dot{x} = 6 \cos 2t + 8 \sin 2t$

$\ddot{x} = -12 \sin 2t + 16 \cos 2t$

$= -4[3 \sin 2t - 4 \cos 2t]$

$= -4x \quad (\text{ie } -n^2x)$

ii) $t=0 \quad \dot{x} = 6 \cos 0 + 8 \sin 0$
 $= 6$

iii) $\dot{x} = 0 \quad 6 \cos 2t + 8 \sin 2t = 0$

$8 \sin 2t = -6 \cos 2t$

$\tan 2t = -\frac{3}{4}$

$2t = \pi - 0.6435$

$t = 1.249$

c) i) $\ddot{x} = -\frac{1}{4x^3}$

$\frac{d}{dx}(\frac{1}{2}v^2) = -\frac{1}{4}x^{-3}$

$\frac{1}{2}v^2 = \frac{1}{8}x^{-2} + C$

$v = \frac{1}{2}x = 1 \quad \frac{1}{2}(\frac{1}{4}) = \frac{1}{8} + C$

$\therefore C = 0$

$\frac{1}{2}v^2 = \frac{1}{8x^2}$

$v = \frac{1}{2x}$

ii) $\frac{dx}{dt} = \frac{1}{2x} \quad \frac{dt}{dx} = 2x$

$t = x^2 + C$

Marks Awarded	Marker's Comments
a) i) 1	Correct expression for $\ddot{x}$
1	Correct manipulation to $\ddot{x} = -n^2x$ (well done)
ii) 1	$\dot{x} = 6$ (Very well done)
iii) 1	Let $\dot{x} = 0$ (Generally OK)
1	$\tan 2t = -3/4$ Some poor solutions do
1	$t = 1.249$ $\tan 2t = -3/4$
b) i) 1	Correct version of $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$
1	Evaluate $C = 0$
1	Correct "tidy up" to $v = \frac{1}{2x}$ (many forgot C)
ii) 1	For $\frac{dt}{dx} = 2x$
1	Evaluate C.
1	Manipulate to $x = \sqrt{t+1}$

MATHEMATICS EXTENSION 1 - QUESTION 7

a) Initially, $\dot{x} = V \cos \theta = 40 \cos 60^\circ = 20$; $\dot{y} = V \sin \theta = 40 \sin 60^\circ = 20\sqrt{3}$

$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\dot{x} = c$$

$$\dot{y} = -10t + k$$

$$\dot{x} = 20$$

$$\text{When } t=0, \dot{y} = 20\sqrt{3} \therefore k = 20\sqrt{3}$$

$$x = 20t + c'$$

$$\therefore \dot{y} = 20\sqrt{3} - 10t$$

$$\text{When } t=0, x=0 \therefore c' = 0$$

$$y = 20\sqrt{3}t - 5t^2 + k'$$

$$\therefore x = 20t \quad (A)$$

$$\text{When } t=0, y=0 \therefore k' = 0$$

$$\therefore y = 20\sqrt{3}t - 5t^2 \quad (B)$$

$$\text{From (A), } t = \frac{x}{20} \therefore y = 20\sqrt{3} \left(\frac{x}{20}\right) - 5 \left(\frac{x}{20}\right)^2$$

$$y = \sqrt{3}x - \frac{x^2}{80}$$

(4)

b) Equation of slope: $y = \frac{1}{4}x$ $\therefore \sqrt{3}x - \frac{x^2}{80} = \frac{1}{4}x$

$$80\sqrt{3}x - x^2 = 20x$$

$$(80\sqrt{3} - 20)x - x^2 = 0$$

$$x[80\sqrt{3} - 20 - x] = 0$$

$$x = 0 \text{ or } x = 80\sqrt{3} - 20$$

$$\therefore \text{Horizontal distance is } 80\sqrt{3} - 20$$

(2)

c) From $y = \frac{1}{4}x$, $y = \frac{1}{4}(80\sqrt{3} - 20) = 29.64$

$$x = 80\sqrt{3} - 20 = 118.56$$

$$\text{Distance OA} = \sqrt{118.56^2 + 29.64^2}$$

$$= 122 \text{ metres (nearest metre)}$$

(2)

d) Height above slope: $H = \sqrt{3}x - \frac{x^2}{80} - \frac{1}{4}x$

$$\frac{dH}{dx} = \sqrt{3} - \frac{1}{4} - \frac{x}{40}$$

$$\frac{dH}{dx} = 0: x = 40(\sqrt{3} - \frac{1}{4})$$

This is for maximum value of H (concave down parabola)

$$\text{When } x = 40(\sqrt{3} - \frac{1}{4}), H = (\sqrt{3} - \frac{1}{4})40(\sqrt{3} - \frac{1}{4}) - \frac{1}{80} \cdot 40^2(\sqrt{3} - \frac{1}{4})^2$$

$$= 40(\sqrt{3} - \frac{1}{4})^2 - 20(\sqrt{3} - \frac{1}{4})^2$$

$$= 20(\sqrt{3} - \frac{1}{4})^2$$

$$= 20(3 - \frac{\sqrt{3}}{2} + \frac{1}{16})$$

$$= 61.25 - 10\sqrt{3}$$

$$\text{Maximum height is } (61.25 - 10\sqrt{3}) \text{ m.}$$

(4)

Mathematics Extension One: Question Number 7

Marker: C3F

Marks Awarded	Marker's Comments
(a) 1 mark. 1 mark. 1 mark. 1 mark.	$\begin{matrix} \dot{x} = 20 \\ \dot{y} = 20\sqrt{3} \end{matrix} > \text{in whatever form and}$ whenever stated in answer $x = 20t \dots \dots$ deriving equation / finding c' $y = 20\sqrt{3}t - 5t^2 \dots$ deriving equation and finding k and k' (mark not awarded if k and k' ignored) $t = \frac{x}{20} \rightarrow y = 20\sqrt{3} \left(\frac{x}{20}\right) - 5 \left(\frac{x}{20}\right)^2$
(b) 1 mark 1 mark.	$\frac{1}{4}x = \sqrt{3}x - \frac{x^2}{80}$ $x = 80\sqrt{3} - 20$ / derived from above.
(c) 1 mark 1 mark	$y = \frac{1}{4}(80\sqrt{3} - 20)$ and $x = 80\sqrt{3} - 20$ distance OA = 122 m.
(d) 1 mark 1 mark 1 mark 1 mark	$H = \sqrt{3}x - \frac{x^2}{80} - \frac{1}{4}x$ $\frac{dH}{dx} =$ $x = 40(\sqrt{3} - \frac{1}{4})$ $H = 61.25 - 10\sqrt{3}$ Note: $t = 2\sqrt{3} - \frac{1}{2}$ Not $t = 2\sqrt{3}$. This is because lamp is on a slope and maximum height is not in middle